Diskrete Mathematik – Woche 6

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https://github.com/tschai-yim/eth-resources
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Relational Mechanics

Problem 1 ⋆ ⋆

Let ρ be a relation on a set A.

1. Prove algebraically that the relation $\rho \circ \hat{\rho}$ is always symmetric.
2. Prove algebraically that if ρ is symmetric, then $\rho \circ \rho$ is symmetric.

Problem 2 ⋆ ⋆ ⋆

Prove that if a relation ρ is symmetric, then its transitive closure ρ^* is also symmetric. (Hint: First prove that $\widehat{\rho^n} = (\hat{\rho})^n$)

Relation Properties

Problem 3 ⋆ ⋆

Let ρ and σ be relations on the set $\mathbb{Z}$.

1. If ρ and σ are transitive, is $\rho \cup \sigma$ always transitive? Prove or provide a minimal counterexample.
2. If ρ and σ are symmetric, is $\rho \cup \sigma$ always symmetric? Prove or provide a minimal counterexample.

Problem 4 ⋆ ⋆

Let ρ be a relation on A.

1. Disprove the claim: "If ρ is irreflexive, then ρ^2 is irreflexive."
2. Prove the claim: "If ρ is irreflexive and transitive, then ρ is antisymmetric."

Problem 5 ⋆ ⋆ (extra)

Let ρ and σ be antisymmetric relations on a set A.

1. Prove that the intersection $\rho \cap \sigma$ is antisymmetric.
2. Disprove that the composition $\rho \circ \sigma$ is always antisymmetric.

Equivalence Relations

Problem 6 ⋆ ⋆

Let $f : A \to B$ be a function. Define a relation on A by $x \underset{f}{\sim} y \Leftrightarrow f(x) = f(y)$.

1. Briefly show this is an equivalence relation.
2. Let $f : \mathbb{Z} \to \mathbb{Z}$ be $f(x) = |x| \bmod 3$. Explicitly list the equivalence classes of the quotient set $\frac{\mathbb{Z}}{\underset{f}{\sim}}$.

Problem 7 ⋆ ⋆ ⋆

Define a relation ρ to be **circular** if $\forall a, b, c : (a \rho b \wedge b \rho c) \Rightarrow c \rho a$. Prove the following theorem: A relation ρ is an equivalence relation if and only if it is both reflexive and circular.

Problem 8 (extra)

Given an arbitrary relation ρ on A. Let $\sigma_1 = \rho \cup \hat{\rho} \cup \mathrm{id}_A$.

1. ($\star$) Is σ_1 always an equivalence relation? Justify your answer.

2. ($\star\star\star$) Express the **smallest** equivalence relation σ_2 containing ρ using union, inverse, identity and transitive closure, and prove that your expression is correct.

Partial Orders

Problem 9 $\star$

We know $(\mathbb{N}; \leq)$ is well-ordered.

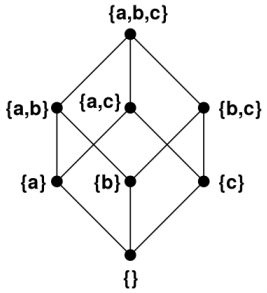
1. Is $(\mathbb{Z}; \leq)$ well-ordered? Justify your answer.
2. Is the subset of rational numbers $S = \{q \in \mathbb{Q} \mid 0 \leq q \leq 1\}$ well-ordered by $\leq$? Justify your answer.

Problem 10 $\star\star$

In any partially ordered set $(A; \preceq)$, let $S \subseteq A$. Formally prove that if S has a least upper bound (supremum), it is unique.

Problem 11 $\star$ (extra)

Consider the Boolean lattice $(\mathcal{P}(\{a, b, c\}); \subseteq)$, which forms a cube in its Hasse diagram (see Figure 3.1 in script). Find a set of 8 integers X such that the poset $(X; |)$ has the exact same Hasse diagram structure.



Problem 12 $\star\star$ (extra)

Let $\left(A; \underset{A}{\preceq}\right)$ and $\left(B; \underset{B}{\preceq}\right)$ be two posets. Recall the direct product relation $\leq$ on $A \times B$ from Definition 3.28: $(a_1, b_1) \leq (a_2, b_2) \Leftrightarrow a_1 \underset{A}{\preceq} a_2 \wedge b_1 \underset{B}{\preceq} b_2$.

Formally prove that $\leq$ is antisymmetric on $A \times B$.

Problem 13 $\star\star\star$ (extra)

Consider the claim: "If a partially ordered set $(A; \preceq)$ has exactly one minimal element m, then m is the least element of A."

1. Prove this claim is true if A is **finite**. (Hint: use a proof by contradiction).
2. Disprove this claim for **infinite** posets by describing a counterexample.